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Microaeration Process Design, Control, and Monitoring for Biogas Desulfurization in Anaerobic Digestion

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ABSTRACT

Microaeration has emerged as an effective in situ approach for biogas desulfurization in anaerobic digestion (AD). This review critically assessed the effects of key microaeration parameters, including injection locations (liquid vs. gas phase), oxygen source (air vs. pure oxygen), dosage, and process control and monitoring techniques. This review identified several challenges to its large-scale implementation, including a lack of optimal process design and limited economic analysis. Other key challenges include balancing oxygen injections to prevent methanogenic inhibition and organic co-oxidation, managing biogas dilution by residual oxygen and nitrogen, and the operational complexities associated with elemental sulfur removal. Proportional–integral–derivative (PID)-controlled systems that utilize oxidation-reduction potential and H_2S feedback have strong potential for process automation; however, real-time monitoring tools and process control strategies are still in early stages of development. Moreover, its impact on hydrolysis, volatile solids reduction, and methane yield has been inconsistent. Addressing these gaps is crucial for the wide-scale application in AD facilities.

1 | Introduction

In recent years, the growing emphasis on sustainability and environmental protection has shifted global attention from conventional waste disposal toward energy- and resource-recovery processes [1]. Notably, anaerobic digestion (AD) has gained widespread application as a biological approach for treating organic waste due to its dual benefits of waste reduction and renewable energy production [2, 3]. AD generates biogas, reduces sludge volume, deactivates pathogens, and yields nutrient-rich digestate for potential agricultural use. The biogas from AD usually composed of approximately 55%–65% methane (CH_4), 35%–45% carbon dioxide (CO_2), and trace levels of other gases, such as hydrogen sulfide (H_2S), ranging from 50 to 5000 ppm, and can be further upgraded to biomethane (also called renewable natural gas, RNG) or used for combined heat and power (CHP) generation [4, 5].

The presence of H_2S in biogas poses significant challenges. H_2S in AD is primarily formed via reduction of sulfate by sulfate-reducing bacteria (SRB) as well as degradation of organosulfur compounds (e.g., proteins) [6]. During biogas combustion, H_2S is converted to sulfur oxides (SO_x), which can form corrosive condensates that damage infrastructure associated with gas engines, gas pipelines, CHP units, boilers, and other biogas upgrading equipment [7]. Depending on the end-use application, H_2S must

Abbreviations: AD, anaerobic digestion; ADM1-Ox, oxygen-enhanced anaerobic digestion model no.1; BRT, biogas residence time; CFD, computational fluid dynamics; CH_4, methane; CHP, combined heat and power; CO_2, carbon dioxide; COD, chemical oxygen demand; DO, dissolved oxygen; EN, European Norm; H_2S, hydrogen sulfide; HRT, hydraulic retention time; LCA, life cycle assessment; NA, not applicable; O_2, oxygen; ORP, oxidation-reduction potential; OTR, oxygen transfer rate; OUR, oxygen utilization rate; PID, proportional–integral–derivative; RNG, renewable natural gas; S^0, elemental sulfur; $S_2O_3^{2-}$, thiosulfate; SCOD, soluble chemical oxygen demand; SEM, scanning electron microscopy; SO^{42-}, sulfate; SOB, sulfide-oxidizing bacteria; SO_x, sulfur oxides; SRB, sulfate reducing bacteria; TEA, techno-economic analysis; TSs, total solids; UASB, upflow anaerobic sludge blanket; VFAs, volatile fatty acids; VOCs, volatile organic compounds; VSs, volatile solids.

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be reduced to below ~1000 ppm for boilers and turbines and below 0.1 ppm for internal combustion engines [2, 8]. For vehicle fuel applications or injection into the natural gas grid, much stricter H_2S limits are applied, typically ≤5 ppm and often ≤1 ppm, depending on regional standards, to prevent corrosion and meet SO_x emission limits [6, 8]. Moreover, some regulatory agencies enforce stricter limits under Best Available Control Technology (BACT) requirements where feasible. Dissolved sulfide, particularly free H_2S, can inhibit methanogens, reducing process stability and methane yields [9].

To address challenges associated with H_2S in biogas, both in situ and ex situ desulfurization methods have been developed. Ex situ methods, such as chemical scrubbing, adsorption, and bio-trickling filters, involve posttreatment of biogas, which can lead to high operational costs, media replacement, and chemical demand [10]. In contrast, in situ approaches directly applied to the digesters to reduce H_2S formation or enhance its biological oxidation, offering operational simplicity and a low-cost solution [7, 11]. Among these strategies, microaeration has emerged as a promising in situ biogas desulfurization technique due to low energy demand, minimal maintenance, and dual chemical–biological reaction pathways [11, 12]. Existing studies have demonstrated that carefully controlled air/oxygen dosing can achieve high H_2S removal efficiencies (often >90%), although reported performances vary widely depending on reactor configuration, dosing strategy, and microbial community response. These variations highlight the critical need to review prior work to identify clear trends and guidelines.

Microaeration involves the controlled injection of small amounts of air or oxygen (O_2), into the digester headspace or liquid phase, promoting partial oxidation of sulfide. However, the enhanced desulfurization efficiency of biogas desulfurization depends on the range of operational parameters. The injection location (liquid vs. headspace), headspace volume, oxygen source (air vs. oxygen), rate of air/oxygen dosing, microaeration intensity, microaeration frequency, oxidation-reduction potential (ORP), and operational temperature (mesophilic vs. thermophilic) are important factors to consider [13]. Furthermore, the residual oxygen in the biogas must meet the requirements of the downstream biogas utilization processes. Therefore, optimal process control and precise air/oxygen supply, consistent with the H_2S concentration in the biogas, are crucial for achieving higher microaeration efficiency while ensuring compatibility with downstream biogas utilization requirements [11].

Microaeration can potentially offer benefits beyond desulfurization. Previous studies have reported enhanced methane yield [14], improved process stability [15], and accelerated hydrolysis [16], particularly for lignocellulosic substrates [17, 18]. Moreover, it can reduce the accumulation of volatile fatty acids (VFAs) [19], accelerate the fermentation process [20], and enhance the degradation of long-chain fatty acids [21], organic micropollutants [22], and polymeric siloxanes [23]. Its positive influence on pH stability and facultative microbial dynamics further enhances the resilience of AD [11, 13]. A recent study reported that microaeration can reduce the prevalence of antibiotic resistance genes and mobile genetic elements, potentially decreasing the risk of spreading antibiotic resistance through land application of digestate [24]. Although these studies collectively indicate the potential of microaeration, their findings are spread across diverse reactor scales and operational settings, complicating direct comparison.

Several existing reviews have broadly discussed microaeration performance; however, most do not provide a comprehensive synthesis that explicitly links operational strategies, process control parameters, and monitoring techniques with the underlying mechanisms and biogas desulfurization efficiency across different reactor scales. Thus, there is a need for a more comprehensive and updated review. This review aims to fill this specific gap by providing a critical, up-to-date assessment of the literature on microaeration-based biogas desulfurization in AD. First, the review summarizes existing data on the application of microaeration in AD, providing a comprehensive overview of various microaeration strategies, including injection locations (liquid vs. gas phase), headspace volumes, and oxygen sources (air vs. oxygen), along with their respective advantages and disadvantages. Second, the review examines the parameters and monitoring techniques of microaeration, including dosage, duration, frequency, and intensity, to improve biogas desulfurization. Moreover, the effects of microaeration on AD performance across different reactor scales are compared. Finally, key research gaps, challenges, and future research directions are highlighted to guide future research.

2 | Mechanisms, Process Design, and Operation

Conventionally, exposing anaerobic digesters to aerobic conditions should be avoided to prevent adverse effects on obligate anaerobes, especially methanogens, which are crucial to AD. However, over the past two decades, controlled dosing of air or pure oxygen in the AD system has gained significant attention for its operational benefits. Microaeration promotes the partial oxidation of sulfides. As shown in Figure 1, the process is primarily mediated by sulfide-oxidizing bacteria (SOB), which convert H_2S into elemental sulfur (S^0), sulfate (SO_4^{2-}), and thiosulfate ($S_2O_3^{2-}$), depending on oxygen availability [11]. Elemental sulfur is the predominant product under oxygen-limited conditions, requiring only 0.5 mol O_2 per mol S^{2-}, a key factor for the cost-effective operation of microaeration [12, 25]. The relevant oxidation reactions are shown in the following equations [10, 14, 24]:

$$2H_2S + O_2 \rightarrow 2S^0 + 2H_2O \quad (1)$$

$$H_2S + 2O_2 \rightarrow SO_4^{2-} + 2H^+ \quad (2)$$

$$2S + 2H_2O + 3O_2 \rightarrow 2SO_4^{2-} + 4H^+ \quad (3)$$

$$2HS^- + O_2 \rightarrow 2S^0 + 2OH^- \quad (4)$$

$$2HS^- + 4O_2 \rightarrow 2SO_4^{2-} + 2H^+ \quad (5)$$

Previous studies have demonstrated that even a small amount of oxygen can effectively remove H_2S from biogas, enhance hydrolysis, and increase microbial diversity, including methanogenic archaea and facultative bacteria, thereby improving digester stability and methane yields [14, 27]. The performance of microaeration-optimized AD can be influenced by operational

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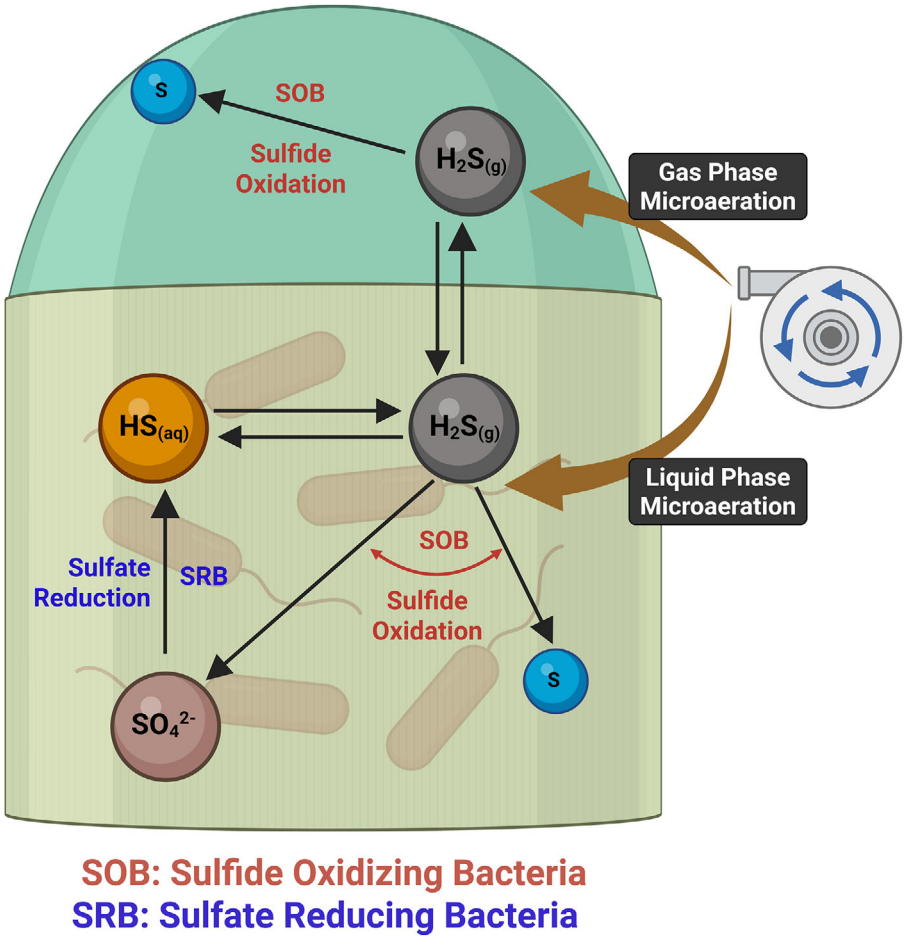


FIGURE 1 | Conceptual schematic of gas-phase and liquid-phase microaeration pathways in anaerobic digestion system. SOB, sulfide-oxidizing bacteria; SRB, sulfate-reducing bacteria.

parameters, among which the location of air or oxygen injection and the oxygen source both play critical roles.

2.1 | Injection Location (Liquid vs. Gas Phase)

A key operational consideration is the choice between gas-phase and liquid-phase microaeration. Microaeration is typically applied either in the liquid phase, by injecting air or oxygen into the sludge recirculation line or directly into the digester liquid, or in the gas phase, by injecting into the biogas recirculation line or into the digester headspace [28]. Figure 2 shows conceptual schemes of liquid-phase and gas-phase microaeration for in situ biogas desulfurization in AD. Gas-phase injections, typically in the headspace, enable direct oxidation of gaseous $H_2S_{(g)}$, thereby reducing oxygen demand and nitrogen dilution of biogas. In contrast, liquid-phase injections are susceptible to oxygen losses due to consumption by facultative bacteria or oxidation of other organics, which can increase aeration requirements and the risk of methanogenic inhibition [29].

Gas-phase microaeration, particularly through air or oxygen injection into the digester headspace, is more extensively reported in the literature (see Table 1). One of the key advantages of headspace injections is that gaseous H_2S readily reacts with oxygen, thereby reducing the oxygen requirement per unit of H_2S removed [30, 31]. This also minimizes the dilution of biogas caused by the addition of nitrogen and oxygen, which is important for maintaining the biogas quality [28, 32]. However, it is essential to control low residual oxygen in biogas to meet process safety and combustion standards. Sousa et al. reported that injection of air into the headspace achieved higher H_2S removal efficiency in microaerobic treatment of low-strength wastewater than injection into the liquid phase, though a slight reduction in methane content occurred due to nitrogen dilution [33]. In another study, Di Costanzo et al. achieved 98.2% ± 1.3% H_2S removal in full-scale digesters using oxygen for headspace microaeration without negatively impacting methane production, although the long-term economic impact of increased operational and maintenance requirements was not fully addressed [5]. Additionally, Pokorna-Krayzelova et al. suggested using a silicon-based biomembrane to enhance H_2S removal under microaerobic conditions while preventing oxygen and nitrogen contamination of biogas [34]. These findings support headspace microaeration as a reliable and effective strategy for biogas desulfurization, offering high H_2S removal efficiency with minimal impact on methane production when properly controlled.

Liquid-phase microaeration, which involves injecting oxygen or air into the digester liquid or the sludge recirculation line, presents different operational challenges. One major challenge is the potential for co-oxidation of organic compounds, which

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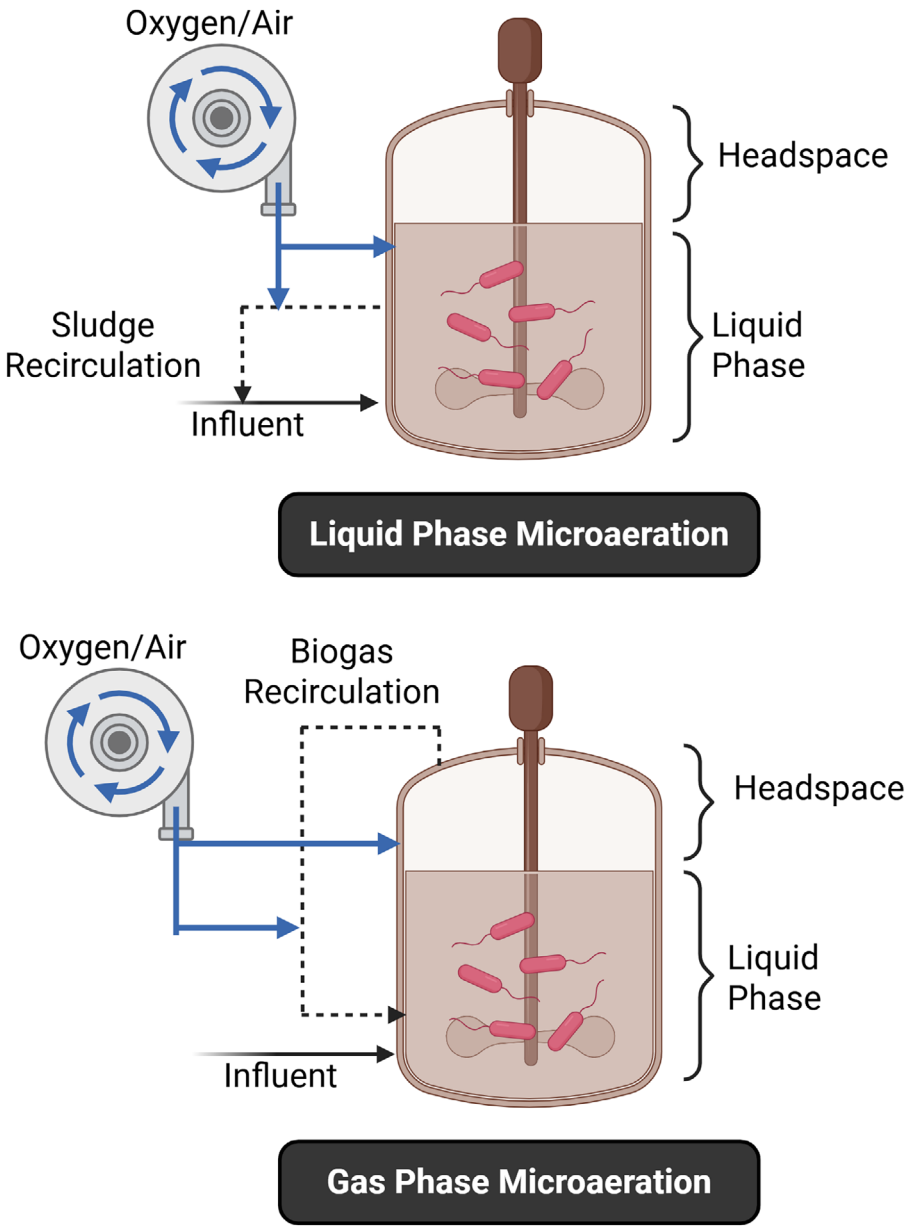


FIGURE 2 | Conceptual schemes for microaeration for in situ H_2S removal in anaerobic digestion: the liquid-phase microaeration and gas-phase microaeration.

can reduce the amount of oxygen available for H_2S removal and can negatively impact methane production. Some studies have reported that 13%–59% of the oxygen injected into the liquid phase was consumed by nonspecific oxidation reactions [35]. For instance, Krayzelova et al. found that only 20%–30% of dissolved sulfide was removed when oxygen was added to the liquid phase, whereas H_2S removal from biogas reached 79%, suggesting that oxidation still occurred in the headspace even when oxygen was dosed into the liquid phase [12]. Liquid-phase dosing may require a higher oxygen flow rate, which can increase operational cost. Munoz et al. proposed that the ratio of oxygen injection to biogas production rate (O_2/biogas production rate) can serve as a useful parameter for optimizing dosing volume [36]. A study of seven full-scale AD facilities in Europe from 2003 to 2005 achieved 90% H_2S removal by air injection into the sludge recirculation line [11]. However, some studies have observed minimal co-oxidation of organic matter under microaerobic conditions. For example, similar chemical oxygen demand (COD) balances were reported in microaerobic and control reactors, suggesting that oxygen consumption by organics may be minimal under certain conditions [12]. Nonetheless, other studies showed that liquid-phase microaeration could lower the dissolved sulfide concentrations and, in turn, reduce sulfide toxicity to methanogens, providing an indirect benefit to process stability [12, 37]. However, there were different views on the impacts of injection locations on H_2S removal efficiency from biogas. Ramos et al. evaluated the performance of microaerobic desulfurization units with oxygen injected at different locations. There was no significant variation in H_2S removal efficiencies in their findings, suggesting that the injection location alone may not be the primary determinant of system effectiveness [38]. Therefore, liquid microaeration may entail higher oxygen demand and higher operational

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TABLE 1 | Overview of anaerobic reactors utilizing microaeration for H_2S removal.

Reactor type	Reactor capacity	Substrate	Microaeration dosing location	Initial H_2S level (ppm)	H_2S removal efficiency (%)	References
Lab scale	15 L	Chicken manure	Headspace	5000–6000	99.7	[40]
Lab scale	20 L	Sludge	Headspace	5800	97	[41]
Lab scale	20 L	Cheese whey solution	Headspace	10 000	90	[42]
Lab scale	2 L	Digested sludge	Liquid phase	555	87	[24]
Lab scale	3.5 L	Low strength wastewater	Liquid and headspace	NA	80	[33]
Lab scale	4.5 L	Animal wastewater	Liquid	336	70.2	[43]
Lab scale	10 L	Sewage sludge	Liquid	NA	99	[44]
Lab scale	11 L	Sewage sludge	Sludge recirculation line	NA	92	[45]
Lab scale	20 L	Cheese whey solution	Headspace	6000	97	[46]
Field pilot	8.5 m^3	Manure	Headspace	2300–4000	99	[47]
Pilot scale	7 m^3	Sewage sludge	Headspace, liquid, and biogas recirculation	500	99	[48]
Pilot scale	50 L	Primary sludge	Liquid	6000	99.5	[49]
Pilot scale	200 L	Sludge	Sludge or biogas recirculation line	9318	99	[50]
Pilot scale	250 L	Sewage sludge	Sludge recirculation line	NA	90	[38]
Pilot scale	250 L	Sewage sludge	Headspace	NA	100	[51]
Pilot scale	250 L	Sewage sludge	Headspace	15811	99	[52]
Pilot scale	256 L	Sewage sludge	Sludge recirculation line	NA	99	[29]
Pilot scale	3 m^3	Dairy manure	Headspace	3500	99	[53]
Full scale	338 m^3	Manure	Headspace	2000–4000	68.2	[26]
Full scale	4500 m^3	Wastewater sludge	Headspace	1200–2500	99	[35]
Full scale	1600–30 000 m^3	Sewage sludge	Sludge recirculation line	771–7580	74.99	[11]
Full scale	2450, 3250, and 3300 m^3	Sewage sludge	Sludge recirculation line	1125–5100	86	[54]
Full scale	4500 m^3	Wastewater sludge	Headspace	NA	98.2	[6]

Abbreviation: NA, not applicable.

costs. However, compared to gas-phase microaeration, liquid-phase microaeration can offer multiple benefits, such as easier retrofitting into existing digesters, reduced sulfur deposition in headspace, and lower residual oxygen levels in the biogas. These advantages were initially reported by Jenicek et al. [11] and have since been confirmed by other studies [35, 39].

In summary, although extensive research has demonstrated the effectiveness of both gas-phase and liquid-phase microaeration for H_2S removal, systematic, comparative evaluations of these strategies under similar operating conditions remain lacking, particularly at full or pilot scale. As summarized in Table 1, most studies reported H_2S removal efficiencies exceeding 90%, regardless of the dosing location. However, the relative advantages of specific liquid-phase configurations, such as sludge recirculation versus direct injection, remain also underexplored. Furthermore, the implications of headspace design (ratio of headspace volume to liquid volume), long-term process stability, organic co-oxidation, and maintenance requirements require further investigation. Future studies should adopt standardized evaluation criteria and include economic assessments to guide the optimal selection of microaeration design strategies. Addressing these gaps will be crucial to selecting optimal dosing approaches and ensuring the practical, long-term application of microaeration in full-scale biogas production systems.

2.2 | Headspace Volume and Biogas Residence Times (BRTs)

In addition to injection location, reactor configuration, particularly headspace volume, is a key factor influencing H_2S removal efficiency, as it affects the gas–liquid interface area available for sulfide oxidation and the physical space for elemental sulfur

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accumulation. Ramos et al. examined the effect of headspace volume on biogas desulfurization in a 266 L pilot-scale digester and found that injecting oxygen into the liquid phase was more effective when sufficient headspace (25 L) was available. In contrast, when the headspace was less than 0.3 L, desulfurization efficiency dropped considerably, even at similar oxygen dosing rates. These results indicate that the available surface area in the headspace plays a critical role in facilitating effective oxidation of H_2S [55]. However, increasing headspace volume can reduce the digester's working volume, potentially affecting its economic feasibility and requiring careful evaluation.

While providing adequate headspace volume appears critical, BRT may be a more practical metric for comparing digesters of different scales or configurations, as well as for standardizing microaeration process design. BRT is defined as the ratio of digester headspace volume to the biogas production rate, representing the time biogas remains in contact with injected O_2 and SOB before exiting the reactor [56]. BRT (in hours) can be expressed in the following equation:

$$\mathrm{BRT} = \frac{24 \times V_{\mathrm{headspace}}}{\mathrm{ABPR}} \tag{6}$$

where $V_{\mathrm{headspace}}$ is the headspace volume (L) and ABPR is the average biogas production rate (L day^{-1}) [56]. Thus, a larger headspace volume or lower biogas flow increases BRT, giving more time for H_2S to be oxidized by SOB. In contrast, higher biogas flow rates shorten the BRT and reduce contact time for H_2S removal. Kraakman et al. also reported that a higher volumetric biogas production rate was associated with lower desulfurization efficiency at a given microaeration dosage [54]. Therefore, incorporating adequate BRT in digesters is essential to provide sufficient headspace volume or manage biogas flow to ensure sufficient retention time, enabling effective sulfide oxidation under microaeration [54].

Elemental sulfur accumulation in the headspace has been observed in several studies [26, 55, 57]. Previous studies noted that the growth of microbial mats consisting mainly of elemental sulfur formed on surfaces such as the digester ceiling, walls, and support structures within the headspace. These mats, which were confirmed by scanning electron microscopy (SEM) to be composed of upward-growing filaments, provided a large specific surface area that may facilitate SOB activity [26, 55, 57]. Although these sulfur mats can function as biofilters, excessive accumulation may lead to clogging, overpressure, and maintenance issues. In contrast, other studies did not observe elemental sulfur accumulation in the headspace. They hypothesized that sulfur may have been discharged with the liquid effluent, although this assumption was not experimentally confirmed [48, 58]. Similarly, Krayzelova et al. reported that only 10% of the generated elemental sulfur was retained in the headspace of an upflow anaerobic sludge blanket (UASB) reactor, with 33% being released into the effluent, likely due to the small volume of the reactor's headspace. Regardless, elemental sulfur accumulation appears to be a widespread concern in digester and pipelines, potentially leading to clogging and operational interruptions [12]. To address this, Ramos et al. recommended periodic cleaning of digesters employing microaeration, suggesting an interval of no more than 14 months [51].

Further research is needed to determine the most suitable cleaning methods and intervals, assess the associated cleaning and maintenance costs, and explore effective operational measures to minimize sulfur deposition on headspace surfaces. In a recent study, Andreides et al. investigated the role of headspace design in biogas desulfurization using a 20 L laboratory digester equipped with three different headspace configurations. Their experimental results, supported by computational fluid dynamics (CFD) simulations, showed that headspace geometry significantly influenced biogas flow, sulfur deposition, and the localization of biofilm formation. These findings suggest that headspace design could be optimized to control the location of elemental sulfur accumulation and improve overall desulfurization efficiency. However, the long-term effectiveness of such design modifications remains unclear and should be further validated in different digester configurations and operational parameters. Additionally, it is crucial to assess the economic feasibility of altering headspace geometry at scale and to examine how varying biogas production rates impact gas flow and sulfur deposition patterns. The adaptability of CFD models to various bioreactor designs also requires more in-depth exploration [46].

2.3 | Oxygen Source (Air vs. Oxygen)

As discussed previously, microaeration can be operated by dosing either air or pure oxygen into the digester's headspace or liquid phase [25]. Although both air and pure oxygen can achieve comparable H_2S removal efficiencies, air is more cost-effective but contains nitrogen, which dilutes the biogas and lowers its calorific value [32, 59]. Despite this, the diluted biogas remains suitable for certain applications, such as engine combustion [60].

Using pure oxygen for microaeration can enhance biogas quality. However, this approach leads to higher operational costs due to the need for specialized equipment for oxygen generation and careful monitoring to prevent overdosing [13]. For instance, Fdz-Polanco et al. achieved over 99% H_2S removal efficiency in a 200 L anaerobic digester by controlling the amount of pure oxygen dosed, thereby maintaining stable biogas production and methane yield [50]. Similarly, Díaz et al. observed comparable H_2S removal efficiencies using either air or pure oxygen at equivalent oxygenation rates [52]. Díaz et al. conducted a techno-economic evaluation comparing pure oxygen, concentrated oxygen (95% O_2), and air across three full-scale digesters. Concentrated oxygen emerged as the most cost-effective, with an operational cost of \$0.0022 per m^3 biogas, followed by air (\$0.0032 per m^3 biogas) and pure oxygen (\$0.0045 per m^3 biogas). The higher cost of pure oxygen was primarily due to the cost of storage and handling infrastructure [61]. These findings align with those of Okoro and Sun [19], who also highlighted the economic advantages of concentrated oxygen [10]. Di Costanzo [6] evaluated three full-scale (4500 m^3 each) thermophilic anaerobic digesters using headspace microaeration. The system achieved stable H_2S removal of 98.2% ± 1.3% with a pure oxygen dose of 0.96 ± 0.03 NL Nm^{-3}. No adverse effects on the digester performance were noted; however, energy demands were higher than those reported for air-based systems. Specifically, pure oxygen injections via a side blower required 4.1 Wh per Nm^3 of biogas produced [6, 35].

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TABLE 2 | Comparative summary of air and oxygen dosing strategies in microaeration applications.

Parameter	Oxygen (O_2) dosing	Air dosing
Operational complexity	Higher (requires dosing control, monitoring, and safety)	Lower (basic equipment required)
Cost	Higher (purification, control, and storage)	Lower
H_2S removal efficiency	Higher (up to 99%)	Higher (up to 99%)
Methane content	Higher (minimal dilution)	Lower (N_2 dilution)
Biogas dilution risk	Lower	Higher (N_2 dilution)
Posttreatment requirements	May require O_2 removal to meet standards	May require N_2 removal for pipeline injection
Best use scenarios	Pipeline injection, high purity biomethane yield	Cost-sensitive operation, general combustion
Process safety consideration	Still requires monitoring, risk of overdosing	Must control to avoid explosive mixture (CH_4 ≥60%, O_2: <12%) [63]
Economic assessment	\$0.0045 per m^3 of biogas Concentrated O_2: \$0.0022 per m^3 of biogas (most economical) [5]	\$0.0032 per m^3 of biogas [61]

Some studies have demonstrated high H_2S removal efficiency with air injection. Jenicek et al. reported a 99% H_2S removal efficiency in full-scale digesters treating sewage sludge under limited air supply, along with a reduction in phosphate, ammonia, soluble chemical oxygen demand (SCOD), and volatile solids (VSs) [45, 62]. Jenicek et al. further investigated seven full-scale (1600–30 000 m^3) anaerobic digesters across central Europe over 12 years, in which air was introduced into the sludge recirculation line [11]. Most digesters achieved over 90% H_2S removal without significant biogas leakage or pipe blockages caused by sulfur deposition. However, higher air dosages were often necessary to compensate for oxygen consumption by other oxidative processes, thereby increasing operational costs. Importantly, the residual oxygen concentration in the biogas remained below the explosive threshold (5%–15% by volume), thereby ensuring operational safety. The methane content decreased by up to 2% due to residual nitrogen, suggesting that pure oxygen may not be essential in all cases [11]. Giordano et al. presented long-term desulfurization data from three full-scale thermophilic digesters using air-based microaeration. Despite achieving 99% H_2S removal, the study highlighted the cost of removing residual O_2 and N_2 to meet the biomethane pipeline standards (European Norm [EN] 16726), which require less than 5 ppm H_2S, 0.001%–1% O_2, and 2.5%–4% CO_2. Although pure oxygen may help meet these standards, its energy demand can increase. Their study reported an electricity usage of 3.4 kWh per t of biogas for air-based microaeration, which is lower than that of some oxygen-based systems [7, 35, 54].

In summary, both air and oxygen serve as practical and effective options for H_2S removal via microaeration. Pure oxygen injection, while enhancing biogas quality, often requires higher energy consumption, specialized equipment, and capital investment. Air is cost-effective and readily available, but it can dilute biogas, typically adding 1%–8% nitrogen, which may reduce combustion performance or necessitate additional upgrading steps. A comparative summary highlighting the advantages, limitations, and implementation challenges of air and pure oxygen injection is provided in Table 2. Maintaining oxygen concentrations between 6% and 12% when methane is ≥60% is essential to avoid explosive conditions. Therefore, conducting site-specific techno-economic assessments is critical for identifying optimal microaeration conditions, including the choice between air and oxygen, and the posttreatment requirements for residual O_2 and N_2. Future research should focus on developing real-time adaptive control systems for oxygen dosing, exploring the long-term feasibility of hybrid air/oxygen strategies, and assessing scalable solutions for gas purification. Establishing clear decision-making frameworks based on digester type, operational goals, and biogas utilization pathways will enhance the safe and efficient application of microaeration technology.

2.4 | Air/Oxygen Dosages

Some potential drawbacks of microaeration include methane dilution (particularly when air is used instead of pure oxygen), gas pipeline clogging due to sulfur accumulation, and possible inhibition of the methanogenesis process. Therefore, it is important to optimize oxygen or air dosage for efficient and stable AD performance [12, 64]. The reported range of air/oxygen dosages in the literature varies significantly, depending on digester configuration, biogas composition, feedstock types, and operational goals. For instance, based on multiple reports from full-scale microaeration demonstrations, microaeration dosages ranged from 0.2 to >8 L O_2 per Nm^3 of biogas [6, 11, 26, 54]. Process efficiency is commonly evaluated using O_2/H_2S supplied-to-consumed ratios, which correlate with conversion products such as sulfate or elemental sulfur. Elemental sulfur is typically the primary end-product at low oxygen concentrations (<0.1 mg L^{-1}) [37], whereas higher O_2/H_2S supply ratios could result in significant sulfate production [65]. Ramos et al. found that increased O_2/H_2S supply ratios did not always enhance oxygen utilization efficiency. Instead, the most efficient results occurred when both O_2 supply and consumption increased, potentially suggesting a need for balanced dosing strategies [38].

Many studies have focused on oxygen availability. Klok et al. highlighted that sulfide concentration could significantly influence the biological oxidation pathways. They found that biological

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oxidation increased at sulfide concentrations of 0–0.15 mmol L^{-1}, decreased between 0.3 and 1.0 mmol L^{-1}, and increased again at concentrations of 1.0–5.0 mmol L^{-1}, indicating complex dynamics that must be considered in dosing control [66]. Díaz et al. explored optimal O_2 injection rates at different H_2S concentrations when oxygen was dosed into the digester headspace. For low H_2S concentrations (0.33% v/v), an O_2 rate of 6.4 NL O_2 per Nm^3 biogas achieved 96% removal, whereas higher H_2S levels (3.38% v/v) required 118 NL O_2 per Nm^3 to reach 99% removal efficiency. They also proposed that the ratio of O_2 dosage to biogas production rate is a reliable parameter for real-time control [32]. Pilot to full-scale studies also support these findings. Ramos et al. showed 100% H_2S removal in a sludge digester with a hydraulic retention time (HRT) of 22–24 days at 0.25–0.30 NL O_2 per L feed, while maintaining oxygen content below 1.03% in the biogas [55]. Kraakman et al. reported >80% H_2S removal at oxygen dosages ranging from 0.26 to 0.70 L O_2 per L feed, corresponding to 0.86–2.04 L O_2 per Nm^3 of biogas [54]. Jeníček et al. reported 99% H_2S removal across seven full-scale digesters using 6.72 L O_2 per Nm^3 of biogas [11]. Similarly, Kobayashi et al. reported a 68.2% reduction in H_2S, with complete desulfurization achieved on some days, using an air injection rate of 20 m^3 day^{-1} [26]. This corresponded to an oxygen addition of about 1% of the biogas production rate, equivalent to 8.7 L O_2 per Nm^3 of biogas. A recent study confirmed these observations under real operational conditions in a centralized AD facility. By gradually reducing the oxygen dosage over 84 days from 0.96 ± 0.03 to 0.19 ± 0.01 NL per Nm^3 biogas, the study found that the removal efficiency declined from 98.2% ± 1.3% to 67.4% ± 0.7%, illustrating a direct relationship between dosage and performance. Importantly, headspace microaeration did not decrease digester stability and maintained biogas composition within biomethane standards [6].

Although various studies have demonstrated the effectiveness of oxygen or air dosing in enhancing biogas desulfurization, a standard framework for determining the optimal dosage range across diverse operational settings remains lacking. Reported O_2/H_2S ratios vary widely due to differences in reactor configuration, sulfide loading rates, and operational goals. This variability suggests the need for comprehensive models that integrate microbial dynamics, sulfide oxidation pathways, and digester-specific characteristics. Additionally, long-term studies are necessary to evaluate the sustainability of O_2 dosing strategies, particularly in full-scale operation, where sulfur accumulation, energy demands, and cost-efficiency are major concerns. Future research should focus on developing practical control systems, improving real-time monitoring tools, and conducting techno-economic evaluations to guide microaeration applications.

2.5 | Microaeration Intensity

The rate of microaeration, also known as intensity, is a key operating factor. Table 3 provides a summary of studies reporting microaeration intensities. This rate can be adjusted depending on specific objectives, such as minimizing VFA accumulation, stabilizing the process, reducing H_2S levels, enhancing VFA production, or improving methane yields. The rate must fall between the minimum threshold required to trigger a biological response and the upper limit beyond which anaerobic conditions prevail [15]. For example, a molar ratio of O_2 to S^{2-} 0.5–1.5 has been used to remove H_2S [11], whereas applying oxygen equivalent to 2%–10% of the substrate's COD can enhance hydrolysis [67]. Although studies have used different approaches to report microaeration intensities, these should be expressed as volumetric oxygen or air dosage normalized by reactor volume and time, enabling direct comparisons across studies with different digester operating conditions. However, as shown in Table 3, across reported studies, microaeration intensities ranged from approximately 0.003 to 6.5 L O_2 per L reactor per day, with substantially lower dosages used for H_2S removal than for hydrolysis enhancement-oriented operation.

It is also important to consider the oxygen transfer rate (OTR) and the oxygen utilization rate (OUR) to ensure an adequate oxygen supply. They are affected by multiple factors, including total solids (TSs) content, reactor configuration, microaeration method, and bubble size [68]. For instance, injecting air or oxygen into the digester headspace may effectively remove up to 99% of H_2S from biogas [25], but it may be insufficient for removing dissolved sulfide from the liquid phase or enhancing hydrolysis due to diffusion limitations at the gas–liquid interface, especially in high-solids AD systems [69, 70].

OTR is affected by factors such as reactor configuration (batch or continuous) and the microaeration approach (type of gas and injection phase), whereas OUR is influenced by the inoculum and the type of feedstock used [15]. Intense or prolonged microaeration may be required for recalcitrant feedstocks, such as lignocellulosic biomass, compared to food wastes or sewage sludge [71–73]. Furthermore, multiple studies emphasized the importance of careful oxygen dosing when enhancing hydrolysis. Excess oxygen can lead to overgrowth of facultative bacteria, which compete with methanogens and reduce the methane yield [74–76]. Understanding the interaction between facultative bacteria and anaerobic methanogenic archaea is essential when optimizing microaeration strategies. It is equally important to study the relationships among microaeration rate, OTR, OUR, and microbial community dynamics. Different microorganisms consume oxygen at different rates, and this variation can affect reactor performance [77]. However, although OTR and OUR are crucial for scaling up lab-scale systems, their roles remain poorly understood. The oxygen-enhanced AD model No. 1 (ADM1-Ox), a modified version of the well-established ADM1, offers a promising tool for simulating the effects of oxygen in AD systems [67, 78]. To address critical research gaps in optimizing microaeration intensity, further studies should integrate real-time oxygen monitoring and dynamic control models (e.g., ADM1-Ox) to establish standardized protocols that can be reliably scaled up. Furthermore, comparative long-term evaluations of microaeration strategies across feedstock types and inoculum are essential for improving performance predictability and economic feasibility in full-scale operations.

3 | Process Control and Monitoring

Maintaining appropriate oxygen levels in the digesters is essential because of the dynamic nature of feed composition and loading rates, which can cause fluctuations in sulfide production. For

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TABLE 3 | Summary of studies reported microaeration intensity (rate) for different applications.

Feedstock	Reactor's capacity	Key objective	Microaeration intensity	Oxygen dosage rate	References
Vinasse	1.7 L	Removal of H_2S	0.7–0.9 L air per L reactor per day (molar ratio of O_2 and S is 8–10)	0.15–0.19 L O_2 per L reactor per day	[37]
Synthetic brewery wastewater	2.7 L	Removal of H_2S	0.5 mol O_2 per S^{2-} (1 L air per day)	0.08 L O_2 per L reactor per day	[12]
Pulp mill wastewater	10.5 L	Removal of H_2S	3–6 mL air per L reactor per min (0.38 mL O_2 per mg S)	0.9–1.8 L O_2 per L reactor per day	[79]
Primary sludge	50 L	Removal of H_2S	0.14 mL O_2 per mg S	NA	[80]
Waste-activated sludge	200 L	Removal of H_2S	0.013–0.024 L O_2 per L reactor per day	0.013–0.024 L O_2 per L reactor per day	[50]
Waste-activated sludge	200 L	Removal of H_2S	0.25 L O_2 per L feed and 1.27 L air per L feed	0.25 L O_2 per L reactor per day	[52]
Waste-activated sludge	200 L	Removal of H_2S	0.25 L O_2 per L sludge	0.01 L O_2 per L reactor per day	[71]
High-solid sewage sludge	4500 m^3	Removal of H_2S	NA	0.96 ± 0.03 NL Nm^{-3} biogas	[6]
Corn straw	0.2 L	Enhance methane yield	12.5, 25, 50, and 100 mL air per L reactor per day	0.003–0.021 L O_2 per L reactor per day	[81]
Food waste and brown water	3 L	Enhance hydrolysis	5 and 7 mL O_2 per L reactor per day	0.005 and 0.007 L O_2 per L reactor per day	[82]
Synthetic food waste	4.6 L	Enhance hydrolysis	129, 258, and 387 L air per kg TS per day	2.1, 4.4, and 6.5 L O_2 per L reactor per day	[76]
Municipal solid waste	260 L	Enhance hydrolysis and acidification	1 L air per min for 2 h and stop 4 h 3.2 L air per kg TS per day	0.39 L O_2 per L reactor per day	[83]
Primary sludge	1.6 L	Enhance hydrolysis	2.5 mL air per min	0.5 L O_2 per L reactor per day	[84]
Primary sludge	0.5 L	Enhance hydrolysis	500 mL air per day	0.21 L O_2 per L reactor per day	[74]
Food waste and brown water	0.25 L	Enhance hydrolysis	0.037 L O_2 per L reactor per day	0.04 L O_2 per L reactor per day	[82]
Waste-activated sludge	200 L	Overcome overloading and improve reactor stability	4.4 NL O_2 per m^3 reactor per day	0.01 L O_2 per L reactor per day	[38]
Waste-activated sludge	10 L	Control VFAs accumulation and improve effluent quality	1.6 L day^{-1}	0.03 L O_2 per L reactor per day	[44]
Grass silage	1 L	Produce VFAs	1 and 4 L air per min	0.5 and 5 L O_2 per L reactor per day	[73]
Synthetic wastewater	3 L	Control pH	0.1 and 1 g O_2 L^{-1} reactor per day at flowrate 0.9 L O_2 min^{-1}	0.09 and 1.9 L O_2 per L reactor per day	[72]

Abbreviations: NA, not applicable; TS, total solid; VFAs, volatile fatty acids.

subsequent biogas utilization, it is critical to ensure that residual oxygen levels comply with technological requirements. For instance, upgrading biogas for vehicle fuel or for injection into the natural gas grid must maintain an oxygen level between 0.5% and 3% post-CO_2 removal [85]. Effective process control requires oxygen dosing to be continuously adjusted in response to changes in H_2S concentration and biogas flow rate.

Several methods have been documented in the literature for quantifying oxygen supply under microaerobic conditions, including

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volume of air/oxygen per reactor volume per day (L per L reactor per day), per unit feedstock (L per L feed), as a percentage of biogas flow (ratio of O_2 to biogas), and as a volumetric flow rate (L day^{-1}). However, there is no standardized protocol for representing microaerobic dosing, making comparison across studies difficult.

Unlike aerobic systems, online dissolved oxygen (DO) measurement is impractical in microaerobic environments due to extremely low oxygen levels (<1 mg L^{-1}) [2, 14]. Moreover, pH shifts are often insufficiently sensitive to reflect metabolic changes. Conventional DO sensors cannot accurately detect DO levels below 0.1 mg L^{-1} [14, 25]. Therefore, real-time, precise oxygen-control strategies are crucial for efficient, stable AD performance.

ORP presents a promising alternative control parameter. ORP reflects the overall electron balance and redox state of the system, providing an indirect but sensitive measure of oxygen availability [14, 15]. Because ORP responds logarithmically to changes in redox conditions associated with oxygen availability, it can indicate very low oxygen levels, including ranges below the practical detection limits of conventional dissolved oxygen sensors. For example, an ORP value of −50 mV corresponds to approximately 0.1 mg L^{-1} DO. ORP values in microaerobic systems typically range from 0 to −300 mV, whereas in strict anaerobic conditions they are typically below −300 mV [14, 15].

Khanal and Huang proposed a feedback-controlled ORP-based dosing strategy, in which oxygen injection is initiated when the ORP drops 10 mV below a predefined threshold and terminated once the ORP increases 10 mV above the target value [86]. Their findings showed that maintaining ORP between −230 and −180 mV effectively removed dissolved and gaseous sulfide while improving methane yield up to 56.3%. Further studies confirmed the effectiveness of ORP for regulating protein metabolism, microbial gene expression, and oxygen dosing for H_2S removal [17, 49]. In parallel, Ramos and Fdz-Polanco applied a control strategy using biogas yield and H_2S concentration to regulate air injection. Biogas production proved effective for systems with steady sulfur loads, whereas H_2S concentration was a more reliable control parameter under fluctuating sulfur inputs [48].

Intermittent aeration strategies, implemented through defined dosing intervals, have demonstrated promising results. For instance, Khongsumran et al. applied 5-min air dosing intervals every 2 h, achieving a maximum cellulose hydrolysis rate of 3.0 mL O_2 per L of reactor per day [87]. Similarly, Jagadabhi et al. used limited oxygen dosing (10 L O_2 per kg VS per min) to boost VFA production in early stage digestion, whereas air injection into leach bed reactors enhanced both VFA production and utilization [73].

Despite ongoing advancements in monitoring techniques, traditional physicochemical parameters (e.g., pH, DO, and ORP) often fall short in accurately forecasting process stability or capturing real-time change in microbial community dynamics. Li et al. [88] suggested that microbial community analysis could complement conventional monitoring approaches, provide anticipatory understanding of process dynamics, and enhance system diagnostics [88]. Additionally, proportional–integral–derivative (PID) controllers have shown promise in automating oxygen injections based on H_2S concentrations and biogas flow rates. In full-scale digesters, PID-controlled air dosing, typically 1%–3% air into the sludge recirculation line, has resulted in up to 90% H_2S removal when H_2S concentration was in the range of 771–7580 ppm [11]. Similarly, PID-regulated headspace injection of 0.2%–2% air, when the H_2S concentration was in the range of 1200–2500 ppm, achieved a 99% reduction in H_2S [35].

In recent years, several studies have demonstrated the effectiveness of PID-based control strategies at various operational scales. PID controllers can be used to manipulate the air/oxygen flow rate to the digester to control the H_2S level in biogas. PID controllers can be set to a desired H_2S level in biogas, serving as the controller's set point. An online sensor can continuously monitor H_2S concentrations in biogas, allowing the controller to measure the difference between the measured value and the set point. On the basis of the error, the controller can generate an output to adjust the air/oxygen flow rate via an air/oxygen injection valve. A schematic of the PID-controlled microaeration system used for H_2S removal in AD is shown in Figure 3.

Nghiem et al. employed a PID-controlled feedback system combined with ORP monitoring in lab-scale digesters, where automated oxygen dosing effectively reduced H_2S concentrations from approximately 6000 to 30 ppm without compromising methane production [49]. Similarly, Jeníček et al. implemented PID-regulated microaeration at full scale, where precise oxygen dosing via sludge recirculation consistently reduced biogas H_2S levels from approximately 7500 to below 100 mg m^{-3} while maintaining stable digester performance [11]. Giordano et al. also reported that headspace aeration controlled via PID regulation achieved near-complete H_2S removal in thermophilic digesters treating high-solids feedstocks. Collectively, these findings highlight the increasing effectiveness of PID controllers in providing precise and dynamic oxygen dosing under fluctuating H_2S levels [35].

In addition to conventional physicochemical indicators, monitoring shifts in microbial communities provides valuable insights into process stability and performance during microaeration. Recent studies have shown that oxygen dosing directly influences the structure and function of microbial communities, particularly by promoting facultative bacteria and SRB that enhance sulfide removal while maintaining methanogenic activity. Fu et al. reported that controlled microaeration shifted microbial aggregates, allowing facultative and aerobic organisms to consume oxygen near the biofilm surface, thereby protecting core methanogens from oxygen toxicity while improving biomass hydrolysis [89]. Such selective enrichment fosters a stable microbial community capable of sustaining efficient digestion even under fluctuating environmental conditions. Although molecular methods such as high-throughput sequencing are not yet practical for real-time process control, they provide valuable baseline data that can inform the design of more robust and adaptive monitoring and control strategies [89].

In summary, precise process control remains fundamental for optimizing microaeration-based AD. ORP-based control has emerged as a promising technique among various strategies due to its sensitive and real-time adaptability. However, reliance on a

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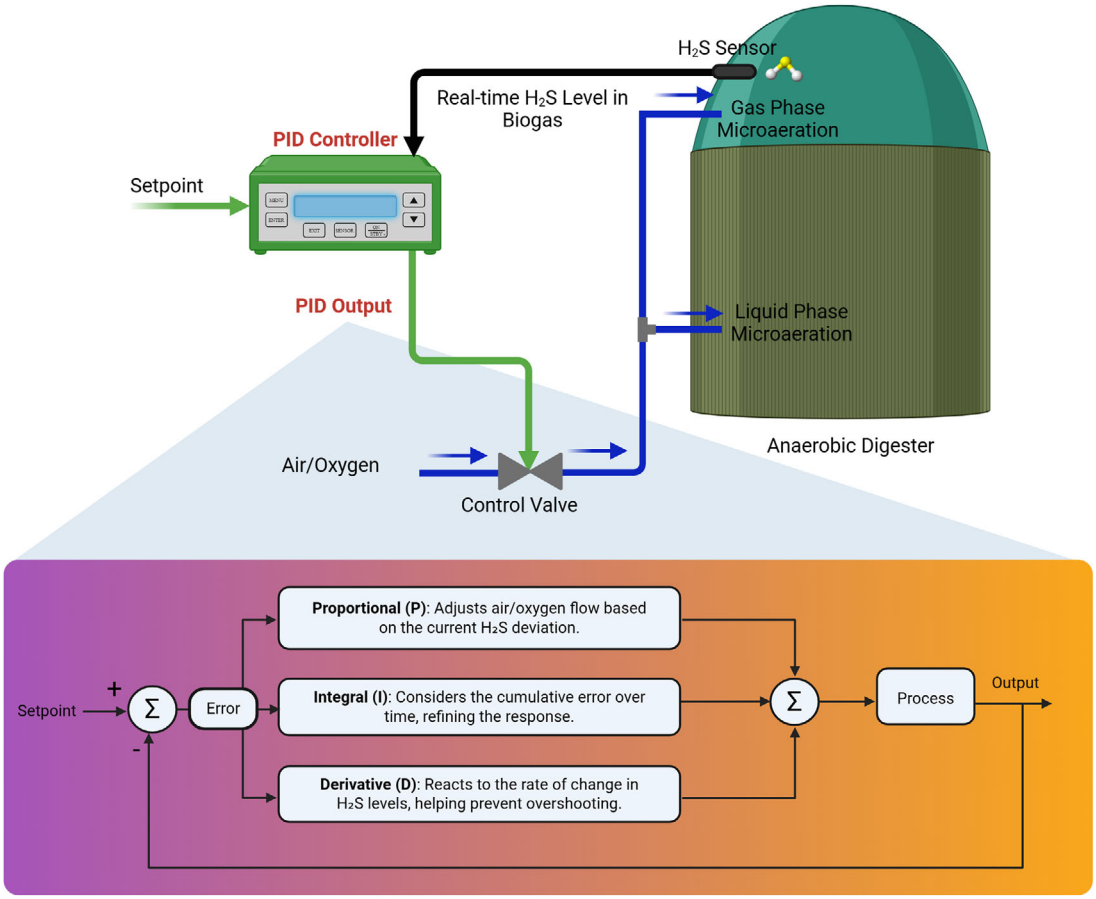


FIGURE 3 | Schematic representation of a PID-controlled microaeration system for H_2S removal in anaerobic digesters. PID, proportional–integral–derivative.

single parameter may be insufficient given the complex dynamic nature of microbial systems. Conventional monitoring parameters such as DO and pH offer limited forecasting capabilities, whereas PID controllers offer automation benefits but require precise calibration. The integration of multiple control strategies, including microbial analysis and data-driven models, could enhance adaptability and operational stability. Future research should focus on standardizing control protocols, enhancing adaptive monitoring technologies, and validating these strategies across diverse feedstocks and full-scale settings.

Figure 4 summarizes various control strategies employed in microaeration-based AD systems, including their control variables, limitations, and advantages. It compares DO monitoring, ORP-based regulations, PID-based automated control systems, and microbial monitoring. Although each method offers specific benefits, such as real-time responsiveness or analytical depth, it also presents challenges related to measurement accuracy, system complexity, and implementation cost.

4 | Process Safety

The safe application of microaeration depends not only on maintaining oxygen concentrations below explosive limits but also on the choice of oxygen dosing strategy. Air is generally considered safer than pure oxygen because of its lower oxygen content (~21%), whereas the use of pure oxygen poses a higher risk of exceeding flammability thresholds if not carefully controlled. In practice, safe operation requires oxygen concentrations in the digester headspace to remain below 12% when methane content in biogas exceeds 60%, ensuring the gas mixture remains outside the explosive range [63]. Site-specific factors, including mixing efficiency, gas residence time, and digester geometry, can significantly influence residual oxygen levels and must be accounted for in system design and operation. Consequently, the use of real-time monitoring and feedback-based control is essential, particularly when applying pure oxygen or high air dosages, to maintain operational safety.

5 | Microaeration Performance at Different Digester Scales

5.1 | Desulfurization Performance

Microaeration has been successfully implemented in different scales of anaerobic digesters. Figure 5 shows the H_2S removal efficiencies reported in lab, pilot, and full-scale studies. The results demonstrate the effectiveness of microaeration across different reactor scales, with most studies reporting over 90% H_2S removal. Although implementing microaeration in large-scale digesters can be challenging, numerous studies have documented successful full-scale applications. Jeníček et al., [11] conducted

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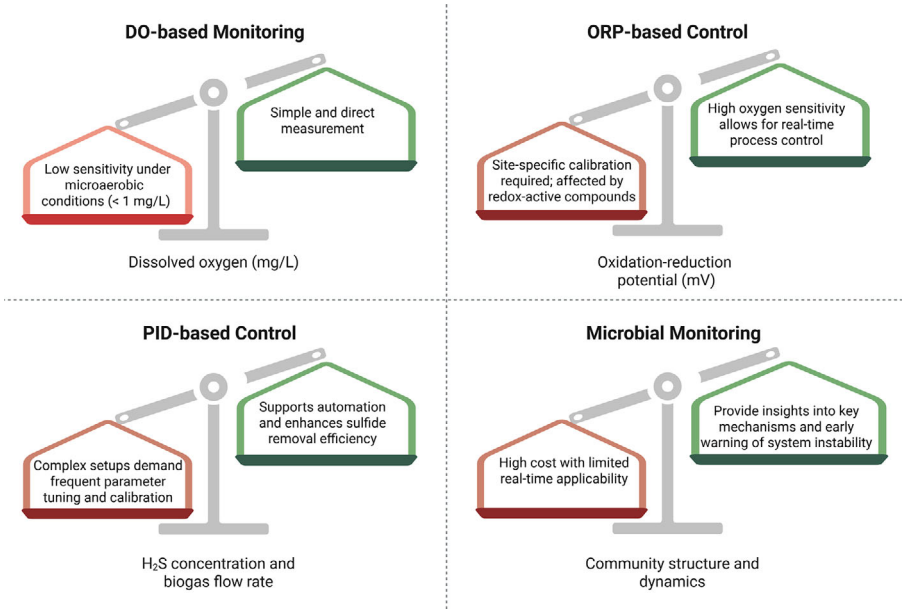


FIGURE 4 | Overview of various process control and monitoring strategies with associated pros and cons in anaerobic digestion under microaeration. DO, dissolved oxygen; ORP, oxidation-reduction potential; PID, proportional–integral–derivative.

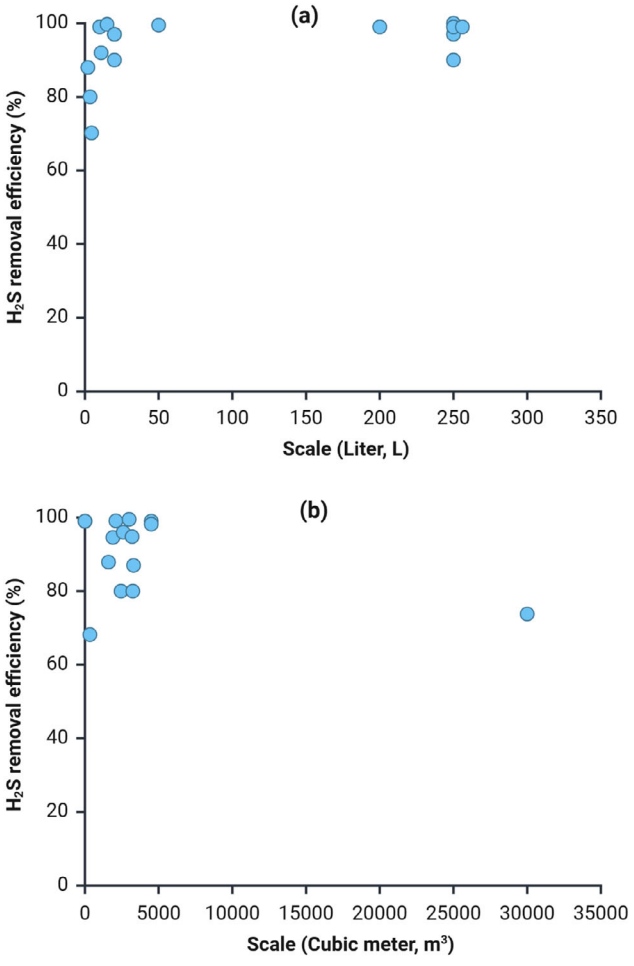


FIGURE 5 | H_2S removal efficiencies achieved by microaeration across different digester scales [6, 11, 24, 26, 29, 33, 35, 38, 40–47, 49, 50, 52–55, 58].

long-term research (2003–2015) on the application of microaeration in seven full-scale anaerobic digesters (volumes ranging from 1600 to 30 000 m^3) in central Europe, achieving over 90% desulfurization efficiency in most of these facilities [11]. Giordano et al. [35] reported that three full-scale thermophilic digesters operated for over 81 weeks with microaeration using air, achieving a significant desulfurization efficiency that removed 99% of H_2S from biogas [35]. However, using air requires the removal of residual oxygen and nitrogen, which increases costs. They suggested that using pure oxygen rather than air might be necessary to meet the European standard EN 16726 for biomethane in pipelines, which sets limits on CO_2, O_2, and H_2S content at 2.5%–4%, 0.001%–1%, and <5 ppm, respectively. The study revealed that oxygen-based microaeration consumes 3.4 kWh per t of biogas, a lower energy cost than previous findings [35, 61]. Kraakman et al. reported that micro-aeration reduced H_2S by 80%–87% in three full-scale anaerobic digesters (2450–3300 m^3) at wastewater treatment plants [54]. Their report suggested that H_2S removal efficiency depends not only on the air injection rate and location but also on digestion activity, as indicated by the volumetric biogas production rate. Higher digestion activities led to lower desulfurization efficiency, indicating the need for adaptive air dosing based on AD performance [54]. Di Costanzo et al. reported up to ~98% H_2S removal in a full-scale thermophilic digester using microaeration (0.96 NL O_2 Nm^{-3} biogas), with no measurable inhibition of methane production [6]. A recent report of an 11-month demonstration of microaeration in the sludge recirculation line of a municipal anaerobic digester (one million gallons; 3800 m^3) in the USA reported effectively reduced H_2S concentrations, but biogas dilution was observed [90]. Replacing air with pure oxygen improved H_2S removal, increased methane content, and limited biogas dilution. Although digester performance remained stable, prolonged operation led to sulfur deposition and localized corrosion, highlighting the need for a maintenance protocol to regularly clean for sulfur.

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Collectively, reports from full-scale demonstrations consistently validate the practical feasibility of microaeration and offer critical insights into design and operational considerations that are not captured in laboratory studies. At the same time, field experience has identified critical challenges that are often not realized in laboratory studies, such as managing sulfur deposition and infrastructure corrosion, maintaining biomethane quality, and ensuring process safety. These insights highlight several directions for future research, including the need for standardized process design guidelines for full-scale systems, adaptive oxygen control based on real-time sensors monitoring of process parameters, and process safety and maintenance protocols.

5.2 | Impact on Hydrolysis, Methane Production, and Process Stability

Although microaeration is mainly used for biogas desulfurization, several studies have also examined its effects on overall AD performance, with mixed results. For instance, Jeníček et al. found that microaeration provided a moderate increase in VSs reduction, while also increasing biogas and methane production by 17% and 16%, respectively [11]. Nghiem et al. reported that microaeration reduced H_2S from 6000 to 30 ppm in a 50 L anaerobic digester [49]. However, VS reduction, COD removal, pH, and alkalinity remained unchanged compared to the control, suggesting that the added oxygen was mainly used by SOB rather than affecting overall process performance. In another study, Azizi et al. reported up to 88% H_2S removal from biogas [24]. However, in their study, both control and microaeration reactors showed comparable performance in terms of biomethane yield, COD removal, and total/VSs degradation, suggesting that air was mainly used for biogas desulfurization. These inconsistencies are likely due to differences in reactor configuration, operating conditions, scale, oxygen dosing strategies, and feedstock characteristics. Thus, site-specific optimization would be needed to realize AD performance enhancement beyond desulfurization.

6 | Future Perspectives and Recommendations

Many studies over the last two decades have demonstrated microaeration as a promising method for biogas desulfurization in AD. Multiple full-scale studies have demonstrated desulfurization efficiencies of up to 99% with well-controlled oxygen or air injection; however, design, control, and economic uncertainties still hinder the widespread implementation of this technology. Notably, there is a lack of standardized process design, adaptive control, and monitoring practices. Both gas-phase and liquid-phase microaeration have shown potential for effective biogas desulfurization, yet the lack of systematic comparative studies complicates the selection process. Although liquid-phase microaeration is relatively easy to implement, retrofitting existing digesters for gas-phase microaeration can involve significant modifications and added costs. On the other hand, liquid-phase microaeration poses a risk of excessive organic oxidation. Optimizing oxygen injection strategies, including dosing rates, timing, frequency, and distribution, is essential for maximizing efficiency while preventing unwanted organic co-oxidation. Single-point air/oxygen injection remains the most widely investigated due to its operational simplicity and low capital cost for retrofitting microaeration in existing anaerobic digesters. However, it can be hypothesized that oxygen distribution, rather than dosage alone, may play a critical role in effective desulfurization of biogas. Therefore, multipoint injection of air/oxygen can potentially improve oxygen transfer efficiency, especially in large-scale or poorly mixed reactors. To date, studies comparing single- and multipoint air/oxygen injection in AD are scarce and warrant further investigation.

Precise oxygen dosage is critical to avoid methanogenesis inhibition, explosion risk, and nitrogen dilution of biogas when air is used for microaeration. Removal of residual oxygen and nitrogen during air-based microaeration increases operational complexity. The effects of dosing frequency compared to dosage amount are inadequately explored. There are also limited studies on the optimal methods and intervals for removing elemental sulfur deposits from digesters [35, 54]. Therefore, thorough economic evaluations that account for both capital and operational costs of aeration systems are essential for informed decision-making. Furthermore, comparative life-cycle assessment (LCA)/techno-economic analysis (TEA) of microaeration versus other competitive technologies (e.g., iron dosing) should be conducted.

Process control and automation through real-time monitoring and supervisory control systems are essential for improving precision and reducing operator burden. However, there have been limited studies on these aspects. Traditional process monitoring parameters, such as DO, exhibit limited sensitivity under microaerobic conditions, making ORP a suitable alternative. PID-controlled microaeration systems, especially those that integrate H_2S feedback, have demonstrated high effectiveness. Future research should focus on developing standardized dosing metrics and integrating automation into control frameworks to enable more robust and adaptive microaeration strategies.

Although microaeration has primarily been studied for H_2S removal, its effects on organosulfur compounds, such as mercaptans, and on volatile organic compounds (VOCs), such as siloxanes, are rarely explored. Only a recent study by Azizi et al. demonstrated that microaeration can achieve up to 88% H_2S removal and moderate siloxane reduction (30%) but showed no significant effect on the removal of mercaptans from biogas [24]. The presence of residual oxygen and nitrogen in biogas has not been widely reported, yet it can be a concern. Thus, further post-cleaning of biogas may still be essential, depending on its end use, and this should be considered in future economic assessments.

Although there is a general consensus on the effectiveness of microaeration for H_2S removal, its impact on other parameters, such as enhanced hydrolysis and fermentation, VSs and COD removal, and methane production, has shown inconsistent results. The oxygen-driven niche separation among SOB, SRB, fermentative bacteria, and methanogens remains ambiguous; consequently, the impact of microaeration on enhanced hydrolysis/fermentation and methanogenesis remains largely speculative and requires further investigation. In particular, comprehensive molecular biology analyses may provide further insights into reactor performance beyond desulfurization. Moreover, kinetic modeling efforts, such as the extensions of the standard ADM1 that incorporate oxygen-driven processes (often referred to as ADM1-Ox), which include sulfide oxidation, sulfate reduction,

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and elemental sulfur generation, can provide further insights into desulfurization dynamics under microaerobic conditions [67]. CFD simulations can also play an important role in optimizing oxygen transfer and mass transfer in both liquid- and gas-phase systems [6, 89]. Despite a handful of full-scale demonstrations, the widespread adoption of microaeration technology will depend on addressing key gaps in process design/standardization, automation, and maintenance requirements. Future development should also focus on better integration of real-time monitoring and control, oxygen-distribution strategies (e.g., multipoint or staged injection), including feedback-based oxygen dosing, to maintain stable microaerobic conditions under changing operating environments. Overall, future advances in reactor design, process control, modeling, and pilot validation are expected to support more reliable and wider use of microaeration for biogas desulfurization.

7 | Conclusions

Microaeration is a promising and cost-effective optimization strategy for H_2S removal in biogas during AD. Injection methods, oxygen source, dosing strategies, and real-time monitoring significantly impact microaeration efficiency. Although headspace injections are generally preferred and mostly studied for reducing oxygen requirements and maintaining biogas quality, liquid-phase microaeration offers additional benefits, with easy retrofitting options. Nevertheless, challenges, such as precise oxygen control, sulfur deposition management, and economic feasibility, limit full-scale application of both approaches. Future research should prioritize standardized design, economic analyses, and enhanced process control to support reliable, large-scale microaeration applications.

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Conflicts of Interest

The authors declare no conflicts of interest.

References

1. Á. Robles, M. V. Ruano, A. Charfi, et al., "A Review on Anaerobic Membrane Bioreactors (AnMBRs) Focused on Modelling and Control Aspects," *Bioresource Technology* 270 (2018): 612–626, https://doi.org/10.1016/j.biortech.2018.09.049.

2. S. M. M. Azizi, B. S. Zakaria, N. Haffiez, P. Niknejad, and B. R. Dhar, "A Critical Review of Prospects and Operational Challenges of Microaeration and Iron Dosing for In-Situ Biogas Desulfurization," *Bioresource Technology Reports* 20 (2022): 101265, https://doi.org/10.1016/j.biteb.2022.101265.

3. F. Fazzino, A. Pedullà, and P. S. Calabrò, "Boosting the Circularity of Waste Management: Pretreated Mature Landfill Leachate Enhances the Anaerobic Digestion of Market Waste," *Biofuel Research Journal* 10, no. 1 (2023): 1764–1773, https://doi.org/10.18331/BRJ2023.10.1.2.

4. P. Zhou, M. N. A. Meshref, and B. R. Dhar, "Optimization of Thermal Hydrolysis Process for Enhancing Anaerobic Digestion in a Wastewater Treatment Plant With Existing Primary Sludge Fermentation," *Bioresource Technology* 321 (2021): 124498, https://doi.org/10.1016/J.BIORTECH.2020.124498.

5. N. Di Costanzo, F. Di Capua, A. Cesaro, M. C. Mascolo, F. Pirozzi, and G. Esposito, "Impact of High-Intensity Static Magnetic Field on Chemical Properties and Anaerobic Digestion of Sewage Sludge," *Waste and Biomass Valorization* 14, no. 8 (2023): 2469–2479, https://doi.org/10.1007/s12649-022-01891-x.

6. N. Di Costanzo, F. Di Capua, A. Cesaro, et al., "Headspace Micro-Oxygenation as a Strategy for Efficient Biogas Desulfurization and Biomethane Generation in a Centralized Sewage Sludge Digestion Plant," *Biomass and Bioenergy* 183 (2024): 107151, https://doi.org/10.1016/j.biombioe.2024.107151.

7. C. Muller, K. Guevarra, A. Summers, et al., "A Review of the Practical Application of Micro-Aeration and Oxygenation for Hydrogen Sulfide Management in Anaerobic Digesters," *Process Safety and Environment Protection* 165 (2022): 126–137, https://doi.org/10.1016/j.psep.2022.07.009.

8. S. Rasi, J. Läntelä, and J. Rintala, "Trace Compounds Affecting Biogas Energy Utilisation—A Review," *Energy Conversation and Management* 52, no. 12 (2011): 3369–3375, https://doi.org/10.1016/J.ENCONMAN.2011.07.005.

9. S. J. W. H. Oude Elferink, A. Visser, L. W. Hulshoff Pol, and A. J. M. Stams, "Sulfate Reduction in Methanogenic Bioreactors," *FEMS Microbiology Review* 15, no. 2–3 (1994): 119–136, https://doi.org/10.1111/J.1574-6976.1994.TB00130.X.

10. O. V. Okoro and Z. Sun, "Desulphurisation of Biogas: A Systematic Qualitative and Economic-Based Quantitative Review of Alternative Strategies," *ChemEngineering* 3, no. 3 (2019): 76, https://doi.org/10.3390/chemengineering3030076.

11. P. Jeníček, J. Horejš, L. Pokorná-Krayzelová, J. Bindzar, and J. Bartáček, "Simple Biogas Desulfurization by Microaeration—Full Scale Experience," *Anaerobe* 46 (2017): 41–45, https://doi.org/10.1016/j.anaerobe.2017.01.002.

12. L. Krayzelova, J. Bartacek, N. Kolesarova, and P. Jenicek, "Microaeration for Hydrogen Sulfide Removal in UASB Reactor," *Bioresource Technology* 172 (2014): 297–302, https://doi.org/10.1016/j.biortech.2014.09.056.

13. B. Khoshnevisan, P. Tsapekos, N. Alfaro, et al., "A Review on Prospects and Challenges of Biological H_2S Removal From Biogas With Focus on Biotrickling Filtration and Microaerobic Desulfurization," *Biofuel Research Journal* 4, no. 4 (2017): 741–750, https://doi.org/10.18331/BRJ2017.4.4.6.

14. Q. Chen, W. Wu, D. Qi, Y. Ding, and Z. Zhao, "Review on Microaeration-Based Anaerobic Digestion: State of the Art, Challenges, and Prospectives," *Science of the Total Environment* 710 (2020): 136388, https://doi.org/10.1016/j.scitotenv.2019.136388.

15. D. Nguyen and S. K. Khanal, "A Little Breath of Fresh Air Into an Anaerobic System: How Microaeration Facilitates Anaerobic Digestion Process," *Biotechnology Advances* 36, no. 7 (2018): 1971–1983, https://doi.org/10.1016/j.biotechadv.2018.08.007.

16. I. Zeb, S. Yousaf, M. Ali, et al., "In-Situ Microaeration of Anaerobic Digester Treating Buffalo Manure for Enhanced Biogas Yield," *Renew Energy* 181 (2022): 843–850, https://doi.org/10.1016/j.renene.2021.09.089.

17. C. G. Liu, C. Xue, Y. H. Lin, and F. W. Bai, "Redox Potential Control and Applications in Microaerobic and Anaerobic Fermentations," *Biotechnology Advances* 31, no. 2 (2013): 257–265, https://doi.org/10.1016/J.BIOTECHADV.2012.11.005.

18. C. Song, W. Li, F. Cai, G. Liu, and C. Chen, "Anaerobic and Microaerobic Pretreatment for Improving Methane Production From Paper Waste in Anaerobic Digestion," *Frontiers in Microbiology* 12 (2021): 688290, https://doi.org/10.3389/FMICB.2021.688290.

21969744, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cben.70046 by Hong Kong University Of, Wiley Online Library on [20/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

19. B. Guo, N. Yu, D. G. Weissbrodt, and Y. Liu, "Effects of Micro-Aeration on Microbial Niches and Antimicrobial Resistances in Blackwater Anaerobic Digesters," *Water Research* 196 (2021): 117035, https://doi.org/10.1016/J.WATRES.2021.117035.

20. G. Sołowski, "Analysis of Biogas Component Production During Anaerobic Digestion of Sour Cabbage in Microaeration Conditions Under Different pH Conditions," *Biomass* 2, no. 1 (2022): 14–26, https://doi.org/10.3390/BIOMASS2010002.

21. M. S. Duarte, S. A. Silva, A. F. Salvador, et al., "Insight Into the Role of Facultative Bacteria Stimulated by Microaeration in Continuous Bioreactors Converting LCFA to Methane," *Environmental Science & Technology* 52, no. 11 (2018): 6497–6507, https://doi.org/10.1021/acs.est.8b00894.

22. J. G. D. S. Do Nascimento, E. V. A. Silva, A. B. Dos Santos, M. E. R. Da Silva, and P. I. M. Firmino, "Microaeration Improves the Removal/Biotransformation of Organic Micropollutants in Anaerobic Wastewater Treatment Systems," *Environmental Research* 198 (2021): 111313, https://doi.org/10.1016/j.envres.2021.111313.

23. A. E. Ortiz-Ardila, B. Díez, C. Celis, P. Jenicek, and R. Labatut, "Microaerobic Conditions in Anaerobic Sludge Promote Changes in Bacterial Composition Favouring Biodegradation of Polymeric Siloxanes," *Environmental Science: Processes & Impacts* 23, no. 8 (2021): 1182–1197, https://doi.org/10.1039/d1em00143d.

24. S. M. Mirsoleimani Azizi, B. S. Zakaria, N. Haffiez, et al., "Bench-Scale Liquid-Phase Microaeration and Iron Dosing Comparison for In Situ Biogas Desulfurization and Antibiotic Resistance Management in Mesophilic Anaerobic Digestion," *ACS ES&T Water* 3, no. 8 (2023): 2458–2470, https://doi.org/10.1021/acsestwater.3c00103.

25. L. Krayzelova, J. Bartacek, I. Díaz, D. Jeison, E. I. P. Volcke, and P. Jenicek, "Microaeration for Hydrogen Sulfide Removal During Anaerobic Treatment: A Review," *Reviews in Environmental Science and Biotechnology* 14, no. 4 (2015): 703–725, https://doi.org/10.1007/s11157-015-9386-2.

26. T. Kobayashi, Y. Y. Li, K. Kubota, H. Harada, T. Maeda, and H. Q. Yu, "Characterization of Sulfide-Oxidizing Microbial Mats Developed Inside a Full-Scale Anaerobic Digester Employing Biological Desulfurization," *Applied Microbiology and Biotechnology* 93, no. 2 (2012): 847–857, https://doi.org/10.1007/S00253-011-3445-6.

27. X. Li, Z. Wang, Y. He, et al., "A Comprehensive Review of the Strategies to Improve Anaerobic Digestion: Their Mechanism and Digestion Performance," *Methane* 3, no. 2 (2024): 227–256, https://doi.org/10.3390/methane3020014.

28. X. Zhang, I. Lawan, and U. A. Danhassan, *Advances in Technologies for In Situ Desulfurization of Biogas*, 1st ed. (Elsevier Inc., 2022).

29. I. Ramos, I. Díaz, and M. Fdz-Polanco, "The Role of the Headspace in Hydrogen Sulfide Removal During Microaerobic Digestion of Sludge," *Water Science and Technology* 66, no. 10 (2012): 2258–2264, https://doi.org/10.2166/WST.2012.457.

30. M. Andreides, L. Pokorná-Krayzelová, J. Bartáček, and P. Jeníč, *Environmental Technologies to Treat Sulfur Pollution: Principles and Engineering*, 2nd ed., ed. P. N. L. Lens, (IWA Publishing, 2020), https://doi.org/10.2166/9781789060966_0345.

31. P. Jenicek, C. A. Celis, J. Koubova, and D. Pokorna, "Comparison of Microbial Activity in Anaerobic and Microaerobic Digesters," *Water Science and Technology* 63, no. 10 (2011): 2244–2249, https://doi.org/10.2166/wst.2011.579.

32. I. Díaz, A. C. Lopes, S. I. Pérez, and M. Fdz-Polanco, "Determination of the Optimal Rate for the Microaerobic Treatment of Several H_2S Concentrations in Biogas From Sludge Digesters," *Water Science and Technology* 64, no. 1 (2011): 233–238, https://doi.org/10.2166/WST.2011.648.

33. M. R. Sousa, C. J. S. Oliveira, J. M. Carneiro, et al., "Evaluation of Different Air Dosing Strategies to Enhance H_2S Removal in Microaerobic Systems Treating Low-Strength Wastewaters," *Environmental Technology (United Kingdom)* 40, no. 28 (2019): 3724–3734, https://doi.org/10.1080/09593330.2018.1487470.

34. L. Pokorna-Krayzelova, K. E. Mampaey, T. P. W. Vannecke, J. Bartacek, P. Jenicek, and E. I. P. Volcke, "Model-Based Optimization of Microaeration for Biogas Desulfurization in UASB Reactors," *Biochemical Engineering Journal* 125 (2017): 171–179, https://doi.org/10.1016/J.BEJ.2017.06.009.

35. A. Giordano, F. Di Capua, G. Esposito, and F. Pirozzi, "Long-Term Biogas Desulfurization Under Different Microaerobic Conditions in Full-Scale Thermophilic Digesters Co-Digesting High-Solid Sewage Sludge," *International Biodeterioration & Biodegradation* 142 (2019): 131–136, https://doi.org/10.1016/j.ibiod.2019.05.017.

36. R. Muñoz, L. Meier, I. Diaz, and D. Jeison, "A Review on the State-of-the-Art of Physical/Chemical and Biological Technologies for Biogas Upgrading," *Reviews in Environmental Science and Biotechnology* 14, no. 4 (2015): 727–759, https://doi.org/10.1007/s11157-015-9379-1.

37. F. P. van der Zee, S. Villaverde, P. A. García, and F. Fdz-Polanco, "Sulfide Removal by Moderate Oxygenation of Anaerobic Sludge Environments," *Bioresource Technology* 98, no. 3 (2007): 518–524, https://doi.org/10.1016/j.biortech.2006.02.011.

38. I. Ramos and M. Fdz-Polanco, "The Potential of Oxygen to Improve the Stability of Anaerobic Reactors During Unbalanced Conditions: Results From a Pilot-Scale Digester Treating Sewage Sludge," *Bioresource Technology* 140 (2013): 80–85, https://doi.org/10.1016/J.BIORTECH.2013.04.066.

39. F. Di Capua, D. Spasiano, A. Giordano, et al., "High-Solid Anaerobic Digestion of Sewage Sludge: Challenges and Opportunities," *Applied Energy* 278 (2020): 115608, https://doi.org/10.1016/j.apenergy.2020.115608.

40. Y. Song, A. Mahdy, Z. Hou, et al., "Air Supplement as a Stimulation Approach for the In Situ Desulfurization and Methanization Enhancement of Anaerobic Digestion of Chicken Manure," *Energy and Fuels* 34, no. 10 (2020): 12606–12615, https://doi.org/10.1021/acs.energyfuels.0c01724.

41. M. Andreides, L. Pokorná-Krayzelová, J. Bartáčková, and J. Bartáček, "Stirring-Based Control Strategy for Microaerobic H_2S Removal in Sequencing Batch Anaerobic Digesters," *Fuel* 306 (2021): 121696, https://doi.org/10.1016/J.FUEL.2021.121696.

42. M. Andreides, L. Pokorná-Krayzelová, J. Říhová Ambrožová, E. I. P. Volcke, and J. Bartáček, "Key Parameters Influencing Hydrogen Sulfide Removal in Microaerobic Sequencing Batch Reactor," *Biochemical Engineering Journal* 168 (2021): 107951, https://doi.org/10.1016/J.BEJ.2021.107951.

43. H. Yang and L. Deng, "Using Air Instead of Biogas for Mixing and Its Effect on Anaerobic Digestion of Animal Wastewater With High Suspended Solids," *Bioresource Technology* 318 (2020): 124047, https://doi.org/10.1016/J.BIORTECH.2020.124047.

44. P. Jenicek, C. A. Celis, L. Krayzelova, N. Anferova, and D. Pokorna, "Improving Products of Anaerobic Sludge Digestion by Microaeration," *Water Science and Technology* 69, no. 4 (2014): 803–809, https://doi.org/10.2166/wst.2013.779.

45. P. Jenicek, F. Keclik, J. Maca, and J. Bindzar, "Use of Microaerobic Conditions for the Improvement of Anaerobic Digestion of Solid Wastes," *Water Science and Technology* 58, no. 7 (2008): 1491–1496, https://doi.org/10.2166/wst.2008.493.

46. M. Andreides, L. Pokorná-Krayzelová, and J. Bartáček, "Importance of Digester's Headspace Geometry for the Efficient H_2S Removal Through Microaeration; Experimental and Simulation Study," *Fuel* 362 (2024): 130900, https://doi.org/10.1016/j.fuel.2024.130900.

47. J. K. Huertas, L. Quipuzco, A. Hassanein, and S. Lansing, "Comparing Hydrogen Sulfide Removal Efficiency in a Field-Scale Digester Using Microaeration and Iron Filters," *Energies* 13, no. 18 (2020): 4793, https://doi.org/10.3390/en13184793.

48. I. Ramos, R. Pérez, M. Reinoso, R. Torio, and M. Fdz-Polanco, "Microaerobic Digestion of Sewage Sludge on an Industrial-Pilot Scale: The Efficiency of Biogas Desulphurisation Under Different Configurations and the Impact of O_2 on the Microbial Communities," *Bioresource*

21969744, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cben.70046 by Hong Kong University Of, Wiley Online Library on [20/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Technology 164 (2014): 338–346, https://doi.org/10.1016/J.BIORTECH.2014.04.109.

49. L. D. Nghiem, P. Manassa, M. Dawson, and S. K. Fitzgerald, "Oxidation Reduction Potential as a Parameter to Regulate Micro-Oxygen Injection Into Anaerobic Digester for Reducing Hydrogen Sulphide Concentration in Biogas," *Bioresource Technology* 173 (2014): 443–447, https://doi.org/10.1016/j.biortech.2014.09.052.

50. M. Fdz-Polanco, I. Díaz, S. I. Pérez, A. C. Lopes, and F. Fdz-Polanco, "Hydrogen Sulphide Removal in the Anaerobic Digestion of Sludge by Micro-Aerobic Processes: Pilot Plant Experience," *Water Science and Technology* 60, no. 12 (2009): 3045–3050, https://doi.org/10.2166/wst.2009.738.

51. I. Ramos, R. Pérez, and M. Fdz-Polanco, "The Headspace of Microaerobic Reactors: Sulphide-Oxidising Population and the Impact of Cleaning on the Efficiency of Biogas Desulphurisation," *Bioresource Technology* 158 (2014): 63–73, https://doi.org/10.1016/J.BIORTECH.2014.02.001.

52. I. Díaz, A. C. Lopes, S. I. Pérez, and M. Fdz-Polanco, "Performance Evaluation of Oxygen, Air and Nitrate for the Microaerobic Removal of Hydrogen Sulphide in Biogas From Sludge Digestion," *Bioresource Technology* 101, no. 20 (2010): 7724–7730, https://doi.org/10.1016/J.BIORTECH.2010.04.062.

53. W. Mulbry, K. Selmer, and S. Lansing, "Effect of Liquid Surface Area on Hydrogen Sulfide Oxidation During Micro-Aeration in Dairy Manure Digesters," *PLoS ONE* 12, no. 10 (2017): e0185738, https://doi.org/10.1371/journal.pone.0185738.

54. N. J. R. Kraakman, I. Diaz, M. Fdz-Polanco, and R. Muñoz, "Large-Scale Micro-Aerobic Digestion Studies at Municipal Water Resource Recovery Facilities for Process-Integrated Biogas Desulfurization," *Journal of Water Process Engineering* 53 (2023): 103643, https://doi.org/10.1016/j.jwpe.2023.103643.

55. I. Ramos, M. Peña, and M. Fdz-Polanco, "Where Does the Removal of H2S From Biogas Occur in Microaerobic Reactors?" *Bioresource Technology* 166 (2014): 151–157, https://doi.org/10.1016/j.biortech.2014.05.058.

56. R. Ruan, J. Cao, C. Li, D. Zheng, and J. Luo, "The Influence of Micro-Oxygen Addition on Desulfurization Performance and Microbial Communities During Waste-Activated Sludge Digestion in a Rusty Scrap Iron-Loaded Anaerobic Digester," *Energies* 10, no. 2 (2017): 258, https://doi.org/10.3390/EN10020258.

57. E. Rodríguez, A. Lopes, M. Fdz-Polanco, A. J. M. Stams, and P. A. García-Encina, "Molecular Analysis of the Biomass of a Fluidized Bed Reactor Treating Synthetic Vinasse at Anaerobic and Micro-Aerobic Conditions," *Applied Microbiology and Biotechnology* 93, no. 5 (2012): 2181–2191, https://doi.org/10.1007/s00253-011-3529-3.

58. I. Díaz, S. I. Pérez, E. M. Ferrero, and M. Fdz-Polanco, "Effect of Oxygen Dosing Point and Mixing on the Microaerobic Removal of Hydrogen Sulphide in Sludge Digesters," *Bioresource Technology* 102, no. 4 (2011): 3768–3775, https://doi.org/10.1016/j.biortech.2010.12.016.

59. L. Guerrero, S. Montalvo, C. Huiliñir, J. L. Campos, A. Barahona, and R. Borja, "Advances in the Biological Removal of Sulphides From Aqueous Phase in Anaerobic Processes: A Review," *Environmental Reviews* 24, no. 1 (2015): 84–100, https://doi.org/10.1139/ER-2015-0046.

60. E. Porpatham, A. Ramesh, and B. Nagalingam, "Investigation on the Effect of Concentration of Methane in Biogas When Used as a Fuel for a Spark Ignition Engine," *Fuel* 87, no. 8–9 (2008): 1651–1659, https://doi.org/10.1016/J.FUEL.2007.08.014.

61. I. Díaz, I. Ramos, and M. Fdz-Polanco, "Economic Analysis of Microaerobic Removal of H_2S From Biogas in Full-Scale Sludge Digesters," *Bioresource Technology* 192 (2015): 280–286, https://doi.org/10.1016/j.biortech.2015.05.048.

62. P. Jenicek, J. Koubova, J. Bindzar, and J. Zabranska, "Advantages of Anaerobic Digestion of Sludge in Microaerobic Conditions," *Water Science and Technology* 62, no. 2 (2010): 427–434, https://doi.org/10.2166/wst.2010.305.

63. T. Shelford, C. Gooch, A. Choudhury, and S. Lansing, *A Technical Reference Guide for Dairy-Derived Biogas Production, Treatment, and Utilization* (Cornell PRO-DAIRY, 2019), https://enst.umd.edu/sites/enst.umd.edu/files/_docs/FarmerbiogashandbookFinal.pdf.

64. O. W. Awe, Y. Zhao, A. Nzihou, D. P. Minh, and N. Lyczko, "A Review of Biogas Utilisation, Purification and Upgrading Technologies," *Waste and Biomass Valorization* 8, no. 2 (2017): 267–283, https://doi.org/10.1007/s12649-016-9826-4.

65. M. Fortuny, J. A. Baeza, X. Gamisans, et al., "Biological Sweetening of Energy Gases Mimics in Biotrickling Filters," *Chemosphere* 71, no. 1 (2008): 10–17, https://doi.org/10.1016/J.CHEMOSPHERE.2007.10.072.

66. J. B. M. Klok, M. de Graaff, P. L. F. van den Bosch, N. C. Boelee, K. J. Keesman, and A. J. H. Janssen, "A Physiologically Based Kinetic Model for Bacterial Sulfide Oxidation," *Water Research* 47, no. 2 (2013): 483–492, https://doi.org/10.1016/J.WATRES.2012.09.021.

67. D. Botheju, B. Lie, and R. Bakke, "Xygen Effects in Anaerobic Digestion—II," *Model Identif Control* 31, no. 2 (2010): 55–65, https://doi.org/10.4173/mic.2010.2.2.

68. F. Garcia-Ochoa and E. Gomez, "Bioreactor Scale-Up and Oxygen Transfer Rate in Microbial Processes: An Overview," *Biotechnology Advances* 27, no. 2 (2009): 153–176, https://doi.org/10.1016/J.BIOTECHADV.2008.10.006.

69. K. Boe, D. J. Batstone, J. P. Steyer, and I. Angelidaki, "State Indicators for Monitoring the Anaerobic Digestion Process," *Water Research* 44, no. 20 (2010): 5973–5980, https://doi.org/10.1016/J.WATRES.2010.07.043.

70. J. P. Sheets, X. Ge, and Y. Li, "Effect of Limited Air Exposure and Comparative Performance Between Thermophilic and Mesophilic Solid-State Anaerobic Digestion of Switchgrass," *Bioresource Technology* 180 (2015): 296–303, https://doi.org/10.1016/j.biortech.2015.01.011.

71. I. Díaz, A. Donoso-Bravo, and M. Fdz-Polanco, "Effect of Microaerobic Conditions on the Degradation Kinetics of Cellulose," *Bioresource Technology* 102, no. 21 (2011): 10139–10142, https://doi.org/10.1016/J.BIORTECH.2011.07.096.

72. C. Sawatdeenarunat, S. Sung, and S. K. Khanal, "Enhanced Volatile Fatty Acids Production During Anaerobic Digestion of Lignocellulosic Biomass via Micro-Oxygenation," *Bioresource Technology* 237 (2017): 139–145, https://doi.org/10.1016/J.BIORTECH.2017.02.029.

73. P. S. Jagadabhi, P. Kaparaju, and J. Rintala, "Effect of Micro-Aeration and Leachate Replacement on COD Solubilization and VFA Production During Mono-Digestion of Grass-Silage in One-Stage Leach-Bed Reactors," *Bioresource Technology* 101, no. 8 (2010): 2818–2824, https://doi.org/10.1016/J.BIORTECH.2009.10.083.

74. J. E. Johansen and R. Bakke, "Enhancing Hydrolysis With Microaeration," *Water Science and Technology* 53, no. 8 (2006): 43–50, https://doi.org/10.2166/WST.2006.234.

75. J. W. Lim and J. Y. Wang, "Enhanced Hydrolysis and Methane Yield by Applying Microaeration Pretreatment to the Anaerobic Co-Digestion of Brown Water and Food Waste," *Waste Management* 33, no. 4 (2013): 813–819, https://doi.org/10.1016/J.WASMAN.2012.11.013.

76. S. Xu, A. Selvam, and J. W. C. Wong, "Optimization of Micro-Aeration Intensity in Acidogenic Reactor of a Two-Phase Anaerobic Digester Treating Food Waste," *Waste Management* 34, no. 2 (2014): 363–369, https://doi.org/10.1016/J.WASMAN.2013.10.038.

77. P. Tsapekos, P. G. Kougias, S. A. Vasileiou, G. Lyberatos, and I. Angelidaki, "Effect of Micro-Aeration and Inoculum Type on the Biodegradation of Lignocellulosic Substrate," *Bioresource Technology* 225 (2017): 246–253, https://doi.org/10.1016/J.BIORTECH.2016.11.081.

78. D. J. Batstone, J. Keller, I. Angelidaki, et al., "The IWA Anaerobic Digestion Model No 1 (ADM1)," *Water Science and Technology* 45, no. 10 (2002): 65–73, https://doi.org/10.2166/WST.2002.0292.

79. W. Zhou, T. Imai, M. Ukita, F. Li, and A. Yuasa, "Effect of Limited Aeration on the Anaerobic Treatment of Evaporator Condensate From a

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Sulfite Pulp Mill," *Chemosphere* 66, no. 5 (2007): 924–929, https://doi.org/10.1016/J.CHEMOSPHERE.2006.06.004.

80. P. M. Ndegwa, L. Wang, and V. K. Vaddella, "Potential Strategies for Process Control and Monitoring of Stabilization of Dairy Wastewaters in Batch Aerobic Treatment Systems," *Process Biochemistry* 42, no. 9 (2007): 1272–1278, https://doi.org/10.1016/J.PROCBIO.2007.06.001.

81. S. F. Fu, F. Wang, X. S. Shi, and R. B. Guo, "Impacts of Microaeration on the Anaerobic Digestion of Corn Straw and the Microbial Community Structure," *Chemical Engineering Journal* 287 (2016): 523–528, https://doi.org/10.1016/j.cej.2015.11.070.

82. J. W. Lim, J. A. Chiam, and J. Y. Wang, "Microbial Community Structure Reveals How Microaeration Improves Fermentation During Anaerobic Co-Digestion of Brown Water and Food Waste," *Bioresource Technology* 171 (2014): 132–138, https://doi.org/10.1016/j.biortech.2014.08.050.

83. P. H. L. Nguyen, P. Kuruparan, and C. Visvanathan, "Anaerobic Digestion of Municipal Solid Waste as a Treatment Prior to Landfill," *Bioresource Technology* 98, no. 2 (2007): 380–387, https://doi.org/10.1016/J.BIORTECH.2005.12.018.

84. J. Diak, B. Örmeci, and K. J. Kennedy, "Effect of Micro-Aeration on Anaerobic Digestion of Primary Sludge Under Septic Tank Conditions," *Bioprocess and Biosystems Engineering* 36, no. 4 (2013): 417–424, https://doi.org/10.1007/S00449-012-0798-X.

85. A. Petersson and A. Wellinger, *Biogas Upgrading Technologies – Developments and Innovations*, IEA Bioenergy Task 37 Report No. 20 (IEA Bioenergy, 2009).

86. S. K. Khanal and J.-C. Huang, "Anaerobic Treatment of High Sulfate Wastewater With Oxygenation to Control Sulfide Toxicity," *Journal of Environmental Engineering* 129, no. 12 (2003): 1104–1111, https://doi.org/10.1061/(ASCE)0733-9372(2003)129:12(1104).

87. O. Khongsumran, P. Intanoo, P. Rangsunvigit, S. Chavadej, and M. Leethochawalit, "Enhancement of Anaerobic Digestion of Cellulosic Fraction in Cassava Production Wastewater by Microaeration," *Chemical Engineering Transactions* 39 (2014): 553–558, https://doi.org/10.3303/CET1439093.

88. L. Li, X. Peng, X. Wang, and D. Wu, "Anaerobic Digestion of Food Waste: A Review Focusing on Process Stability," *Bioresource Technology* 248 (2018): 20–28, https://doi.org/10.1016/J.BIORTECH.2017.07.012.

89. S. Fu, S. Lian, I. Angelidaki, and R. Guo, "Micro-Aeration: An Attractive Strategy to Facilitate Anaerobic Digestion," *Trends in Biotechnology* 41, no. 5 (2023): 714–726, https://doi.org/10.1016/j.tibtech.2022.09.008.

90. M. Seib, "Digester Microaeration: A Comprehensive Full-Scale Case Study," in *Proceedings of the Water Environment Federation 2025* (Water Environment Federation,2025), https://doi.org/10.2175/193864718825159749.

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